

# Saadbin Khan

## Contact Information

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🌐 [Personal Website](#)    📄 [Google Scholar](#)    📄 [ResearchGate](#)    🔗 [LinkedIn](#)

## Education

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- 2021 – Present    📄 **Ph.D. in Engineering Mechanics** (CGPA: 3.89, Expected graduation: May 2026)  
**Virginia Tech**  
Dissertation: The role of hydrodynamic slip and fine tracheal geometry in insect respiratory flow.  
Advisor: [Dr. Anne E Staples](#).  
Co-advisor: [Dr. Jake Socha](#).
- 2017 – 2019    📄 **M.S. in Mechanical and Aerospace Engineering**  
**Oklahoma State University**  
Thesis: Near wall structure and deviation from equilibrium in wavy channel turbulence. 🔗  
Advisor: [Dr. Balaji Jayaraman](#).
- 2011 – 2016    📄 **B.S. in Mechanical Engineering**  
**Bangladesh University of Engineering and Technology**  
Thesis: Mixed convection & MHD effect analysis in a lid-driven cavity with sinusoidally curved bottom wall using CNT-water nanofluid in a porous medium. 🔗  
Advisor: [Dr. Mohammad Arif Hasan Mamun](#).

## Research Focus

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- 📄 **Computational Fluid Dynamics, Data Driven Modeling.**
- 📄 **Multiscale Heat and Mass Transfer.**
- 📄 **Numerical Methods, High-Performance Computing.**

## Publications and Presentations

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### Journal Articles

- 1 **Khan, S.** and Staples, A. E. (2026). ‘A tidal Murray’s law framework inspired by insect respiration’. In: *Nature Physics (Ready to submit)*.
- 2 Kenny, M., Stremmer, M., Miller, L., **Khan, S.**, Staples, A. E. and Socha, J. (2026). ‘Does Murray’s law apply to the tracheal system in insects? A 3D study of the beetle *Platynus decentis*’. In: *Journal of Morphology (In review)*.
- 3 **Khan, S.** and Staples, A. E. (2025). ‘Mechanisms of insect respiration 🔗’. In: *Nature Reviews Physics* 7.3, pp. 1–14.
- 4 Jayaraman, B. and **Khan, S.** (2020). ‘Direct numerical simulation of turbulence over two-dimensional waves 🔗’. In: *AIP Advances* 10.2, p. 025034.
- 5 **Khan, S.** and Jayaraman, B. (2019). ‘Statistical structure and deviations from equilibrium in wavy channel turbulence 🔗’. In: *Fluids* 4.3, p. 161.

### Book Chapters

- 1 **Khan, S.** and Hossain, M. K. (2022). ‘Classification and properties of nanoparticles 🔗’. In: *Nanoparticle-based polymer composites*. Elsevier, pp. 15–54.

## Conference Proceedings

- 1 **Khan, S.** and Staples, A. E. (2025). ‘A modified Murray’s law for insect respiration’. In: *Bulletin of the American Physical Society*. Houston, TX, USA: APS.
- 2 **Khan, S.**, Socha, J., Adjerid, K. and Staples, A. E. (2024). ‘Respiratory airflow driven by propagative collapse in insect tracheae ’. In: *Bulletin of the American Physical Society*. Salt Lake City, UT, USA: APS.
- 3 **Khan, S.**, Wilmsen, S., Zaslavsky, A., Dhingra, M., Socha, J. and Staples, A. E. (2023). ‘Influence of tracheal microstructure on insect respiratory flows ’. In: *Bulletin of the American Physical Society*. Washington, DC, USA: APS.
- 4 **Khan, S.**, Dhingra, M., Socha, J. and Staples, A. E. (2022). ‘Effects of hydrodynamic slip and taenidial structure in insect tracheal flows ’. In: *Bulletin of the American Physical Society*. Indianapolis, IN, USA: APS.
- 5 Staples, A. E., **Khan, S.** and Dhingra, M. (2022). ‘Effects of tracheal microstructures on insect respiratory flows ’. In: *Bulletin of the American Physical Society*. University, MS, USA: APS.
- 6 Khan, M. I., Rabbi, K. M., **Khan, S.** and Mamun, M. (2016). ‘Mixed convection analysis in lid-driven cavity with sinusoidally curved bottom wall using CNT-water nanofluid ’. In: *AIP Conference Proceedings*. Vol. 1754. 1. AIP Publishing LLC, p. 040015.

## Presentations

- 1 **Khan, S.**, Socha, J., Adjerid, K. and Staples, A. E. (2024). ‘Respiratory airflow driven by propagative collapse in insect tracheal network’. In: *2024 Fall Fluid Mechanics Symposium*. Blacksburg, VA, USA: Virginia Tech.
- 2 **Khan, S.**, Dhingra, M., Socha, J. and Staples, A. E. (2023). ‘Effects of hydrodynamic slip and taenidial structure in insect tracheal flows’. In: *2023 Engineering Mechanics Research Symposium*. Blacksburg, VA, USA: Virginia Tech.
- 3 **Khan, S.** (2023). ‘Influence of tracheal microstructure on insect respiratory flows’. In: *Fall 2023 Engineering Mechanics Research Seminar Series*. Blacksburg, VA, USA: Virginia Tech.
- 4 **Khan, S.**, Wilmsen, S., Zaslavsky, A., Dhingra, M., Socha, J. and Staples, A. E. (2023). ‘Influence of tracheal microstructure on insect respiratory flows’. In: *Southeast Regional SICB Meeting 2023*. Blacksburg, VA, USA: Virginia Tech.
- 5 **Khan, S.** and Jayaraman, B. (2018). ‘Assessment of atmospheric plume source inversion using sparse reconstruction and Gaussian dispersion ’. In: *2nd Annual OSU MAE Graduate Research Symposium*. Stillwater, OK, USA: Oklahoma State University.

## Awards

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|-------------|---------------------------------------------------------------------------------------------------------------------------|
| 2026        | ■ <b>Alice &amp; Dan Pletta Scholarship</b> , Engineering Mechanics Graduate Program, Virginia Tech, Blacksburg, VA, USA. |
| 2021 – 2022 | ■ <b>BIOTRANS Fellowship</b> , Interdisciplinary Graduate Education Programs, Virginia Tech, Blacksburg, VA, USA.         |
| 2008 & 2010 | ■ <b>Champion</b> , 6th and 8th Bangladesh Mathematical Olympiad.                                                         |

## Teaching Experience

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- 2025     ■ **Instructor of Record** in Department of Mechanical Engineering, Virginia Tech, Blacksburg, VA, USA.
- **Course taught:** Thermodynamics ([ME-2134](#)).
- 2024 – Present     ■ **Teaching Assistant** in Department of Mechanical Engineering, Virginia Tech, Blacksburg, VA, USA.
- **Courses:** Thermodynamics (ME-2134).
- 2021 – 2024     ■ **Teaching Assistant** in Department of Biomedical Engineering and Mechanics, Virginia Tech, Blacksburg, VA, USA.
- **Courses:** Fluid Mechanics: Control Volumes (ESM-3234), Dynamics (ESM-2304), and Engineering Analysis of Physiological Systems (ESM-4106).
- 2017 – 2019     ■ **Teaching Assistant** in Department of Mechanical and Aerospace Engineering, Oklahoma State University, Stillwater, OK, USA.
- **Courses:** Engineering Analysis & Methods (MAE-3013) and Thermodynamics II (MAE-3223).

## Research Experience

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- 2022 – Present     ■ **Research Assistant** in Laboratory for Fluid Dynamics in Nature (FINLAB), Virginia Tech, Blacksburg, VA, USA.
- **PhD project:** The role of hydrodynamic slip and fine tracheal geometry in insect respiratory flow. [[Article](#) (Cover article of *Nat. Rev. Physics*, March, '25)][[Portfolio](#)]
  - PI: Dr. Anne E Staples.
  - Funded by: National Science Foundation (Award no: 2014181).
  - **Role:** Performed numerical simulations of insect respiratory flow using OpenFOAM and OpenLB (Lattice Boltzmann); investigated flow and transport behavior arising from fine tracheal geometry and taenidial structures; analyzed optimal tracheal branching to derive microscale-informed modifications to Murray's law applicable to biological transport networks.
- 2022     ■ **Research Fellow** in STRETCH Lab, BIOTRANS, Virginia Tech, Blacksburg, VA, USA.
- **Project:** Waviness quantification of collagen fibers from simple harmonic generation (SHG) images. [[Portfolio](#)]
  - PI: Dr. Raffaella De Vita.
  - Funded by: Interdisciplinary Graduate Education Programs at Virginia Tech.
  - **Role:** Quantified collagen fiber waviness in SHG image stacks of uterosacral ligaments using straightness and orientation metrics; analyzed multiple samples with the CT-FIRE software; evaluated algorithm performance and failure modes for highly wavy and non-uniform fibers.
- 2021     ■ **Research Fellow** in Socha Lab, BIOTRANS, Virginia Tech, Blacksburg, VA, USA.
- **Project:** Assembly and operational standardization of the Leica RM2125RTS rotary microtome. [[Portfolio](#)]
  - PI: Dr. Jake Socha.
  - Funded by: Interdisciplinary Graduate Education Programs at VT.
  - **Role:** Assembled and operationalized the microtome at Socha Lab; trained on operation; contributed to fixation and embedding protocols through consultation with external domain experts; documented procedures for repeatable tissue sectioning.

## Research Experience (continued)

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- 2017 – 2019    ■ **Research Assistant** in Computational & Flow Science Research (CFSR) Laboratory, Oklahoma State University, Stillwater, OK, USA.
- **MS Project:** Direct numerical simulations (DNS) of wavy channel flow to understand the near-wall turbulence structure. [Articles: [1](#), [2](#)] [[Portfolio](#)]
  - PI: Dr. Balaji Jayaraman.
  - Funded by: NASA Oklahoma EPSCoR.
  - **Role:** Implemented and modified Fortran subroutines in the [Incompact3d](#) framework for non-flat surface flows using immersed boundary method; developed Python post-processing pipelines; performed data-driven analysis for LES-relevant turbulence modeling.
- 2018    ■ **Research Assistant** in Department of Mechanical and Aerospace Engineering, Oklahoma State University, Stillwater, OK, USA.
- **Project:** An improved plume source inversion model for efficient remote leak detection in atmospheric dispersion applications. [[Portfolio](#)]
  - PI: Dr. Jamey Jacob.
  - Funded by: Baker Hughes General Electric.
  - **Role:** Implemented a Python-based forward atmospheric dispersion model using 3D unsteady advection–diffusion equations to improve upon traditional Gaussian dispersion models; developed a sparse-reconstruction-based plume source inversion model in Python for remote leak detection using limited concentration data.

## Service Experience

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- 2023 – Present    ■ **Student Administrative Assistant** in Biological Transport (BIOTRANS) program, Virginia Tech, Blacksburg, VA, USA.
- Role: Organization of semester events, preparation of annual reports, and management of the program website.
- **Journal Reviewer** for *Fluids*, MDPI; *Partial Differential Equations in Applied Mathematics*, Elsevier.
- 2024 – 2026    ■ **Webmaster** in Graduate Engineering Mechanics Society (GEMS) Leadership Team, Virginia Tech, Blacksburg, VA, USA.
- Role: Managing the website.
- 2018 – 2019    ■ **Public Outreach Manager** in Bangladeshi Students Association, Oklahoma State University, Stillwater, OK, USA.
- Role: Outreach and communication, including sharing organizational updates with current students and engaging prospective students from Bangladesh.
- 2016    ■ **Engineering Intern** in Bangladesh Atomic Energy Commission, Dhaka, Bangladesh.
- Role: Inspection of a 3 MW nuclear research reactor, focusing on structural safety, control rod operation, and cooling water regulation.

## Technical Skills

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|---------------------------|-----------------------------------------------------------------------------------------------------------|
| Programming               | ■ Python, MATLAB, Fortran, C++, C, Open MPI, Mathematica.                                                 |
| Platforms                 | ■ Linux/Unix, Windows, HPC clusters.                                                                      |
| Modeling & Analysis Tools | ■ SolidWorks, OpenFOAM, COMSOL, ANSYS Fluent, OpenLB, DSMC, Open MPI, Paraview, ImageJ, Tecplot, Tracker. |
| Experimental Equipments   | ■ Leica RM2125RTS rotary microtome.                                                                       |
| Miscellaneous             | ■ LaTeX, Microsoft Office, Beamer, Inkscape.                                                              |

## Course Projects and Workshops

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### Course Projects

- 2022    ■ “Approximate Solution of 2D Poisson Equation Using Finite Element Method in an Elliptic Domain”, for ESM 5734: Introduction to the Finite Element Method (VT). [[Report](#)]
- 2019    ■ “Optimization of a reparable multi-state system”, for MATH 4553: Linear and Nonlinear Programming (OSU). [[Report](#)]
- 2018    ■ “2-D incompressible Navier-Stokes equation solver for a channel flow”, for MAE 6263: Computational Fluid Dynamics (OSU). [[Report](#)]
- 2017    ■ “Model reduction for 1D convection-diffusion equation”, for MATH 5543: Numerical Analysis of Differential Equation (OSU). [[Report](#)]
- 2014    ■ “Autonomous robot serving as waiter in restaurant by following lines & avoiding obstacles”, for ME 362: Instrumentation and Measurement Sessional (BUET). [[Report](#)]

### Workshops

- 2018    ■ Message Passing Interface (MPI) Workshop Organized by Extreme Science and Engineering Discovery Environment (XSEDE) along with the Pittsburgh Supercomputing Center, Oklahoma State University, Stillwater, OK.
- 2012    ■ AVR microcontrollers and developing autonomous robot workshop presented by Indian Institute of Technology (IIT) in association with “Smartbotics” and organized by IEEE, BUET, Bangladesh.

## Key Courses

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|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mechanics         | ■ Continuum Mechanics, Fluid Mechanics, Solid Mechanics, Fluidics, Intermediate Dynamics, Turbulent Fluid Dynamics, Computational Fluid Dynamics.                                                                                                                          |
| Numerical Methods | ■ Finite Element Method, Numerical Analysis of Differential Equation, Numerical Engineering Analysis.                                                                                                                                                                      |
| Computing         | ■ Linear & Non-linear Programming, Applied Numerical Computing.                                                                                                                                                                                                            |
| Engineering       | ■ Renewable Energy, Power Plant Engineering, Heat Transfer, Thermodynamics, Internal Combustion Engines, Mechanical Engineering Drawing, Refrigeration and Building Mechanical Systems, Machine Design, Fluid Machinery, Machine Tools, Control Engineering, Mechatronics. |

## References

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### Omer San

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### Jake Socha

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Professor, Virginia Tech,  
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### Scott Huxtable

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Associate Professor, Virginia Tech  
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